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## Coordination in Organizations: *An Integrative Perspective*

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### Abstract

Coordination, the process of interaction that integrates a collective set of interdependent tasks, is a central purpose of organizations. In this review we begin by discussing the origins of interest in coordination, tracing some of the classic perspectives. We present a review of recent literature on coordination in organizations arranged according to the mechanisms that help achieve it. We then go beyond this review to provide a framework to understand what different coordination mechanisms and activities accomplish. We propose that coordination mechanisms (such as routines, meetings, plans, and schedules) impact the work of organizations by creating three integrative conditions for coordinated activity: accountability, predictability, and common understanding. We end by examining the implications of such a perspective for future research on coordination in organizations.

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## Introduction

After many years of limited activity, the last two decades have seen a resurgence of interest in the topic of coordination in organizations. As a consequence, we now have a vibrant and rich body of knowledge about how people collectively perform their interdependent work. These findings usefully update our image of coordination to reflect newer forms of work and organizing. However, the new wave of research spans multiple disciplines and pays attention to different aspects of the problem of coordination, leaving us without a coherent framework. It is time to bring the wide-ranging new understandings of coordination back into the fold of organizational theory by taking stock of this area.

Our literature review has three main goals. We begin by presenting a brief historical account of early perspectives on coordination in organizations including the work of Frederick Taylor, Henry Fayol, and of organizational design scholars. Although some of this early research has been disregarded as quaint and outdated, it is in fact quite relevant to our understanding of coordination and organizations. The coordination practices (such as the use of time as an organizing structure) and concepts proposed by early theorists (such as specialization and management control) continue to frame many of the concerns addressed in contemporary research. Moreover, much of the recent literature is a response to the shortcomings of these historical perspectives, and so a reminder of this research establishes the necessary context for understanding the complementary role of contemporary treatments of coordination in organizations. After touching on the historical highlights, we proceed with our review of recent research by showcasing its unique contributions to our understanding of coordination. Finally, after acknowledging some of the limits and challenges in this most recent wave of research, we offer a perspective to frame coordination research in a more general manner and to help advance our understanding of coordination in organizations.

As we delved into the literature for this review, we saw a need to establish some limits to our work. For example, we found that the term “coordination” was used ubiquitously and loosely to treat a variety of phenomena. In some cases coordination is embedded in the treatment of other constructs, such as collaboration and cooperation. And while collaboration and cooperation are indeed indispensable in coordinated activity (Smith, Carroll, & Ashford, 1995), they are conceptually separate ideas. Thus, to delineate a boundary for our review we began by focusing our attention on research that explains how collective work is accomplished within organizations. This research is closely grounded in work settings and seeks to explain the ways in which collective action on a task takes place. In a few instances we also extended the boundary to include research that provided important knowledge on how or why coordination occurs.

Bounding our review in this way also leads us to exclude other literatures that, although concerned with coordination, do not directly focus on the work of organizations. For example, we exclude the extensive literature on coordination in experimental economics (Davis & Holt, 1993), research focused on coordination between organizations or across organizational divisions (Sinha & Van de Ven, 2005), and research that explores how markets can assist in coordination of action (Williamson, 2005). Although these literatures touch on related questions about coordination, the conditions under which these types of coordination take place and the assumptions made in the research are substantially different from coordination within a single organization that is focused on performing its work.

We therefore begin with a look back at the emergence of coordination demands in the railroads of the nineteenth century, some of the first large-scale organizations in modern times (Chandler, 1962). We then describe the role that scientific management played in early thinking about coordination (Fayol, 1949; Taylor, 1916) and the evolution of work arrangements (Jacoby, 1984), followed by a description of the ideas proposed by organizational design scholars (Hickson, Pugh, & Pheysey 1969; Thompson, 1967; Woodward, 1970). In explaining these perspectives we focus on the contributions they have made to our understanding of coordination in organizations, as well as their limitations. After presenting the historical background which frames modern research on coordination in organizations, we proceed with our review of the current literature.

### **Historical Background on Coordination**

The advent of railroads in the eighteenth century engendered considerable interest in coordination. Chandler (1962, p. 23) argued that railroads were the first of a new breed of organization due to their size and complexity. Their emergence and growth created a pressing need for large-scale coordination because railroads needed to ensure that practical concerns, such as loading passengers, delivering cargo, and avoiding crashes with other trains, could be accomplished. The mechanism that enabled railroads to coordinate on a large scale was the printed timetable (Stover, 1970). Timetables, for example, allowed a train to “own” pieces of track at a specific time, which prevented collisions. Through the use of timetables the railroads could coordinate passengers, cargo, and other trains to meet a specific train when it was at a station.

Timetables could only be used to coordinate trains, though, thanks to the invention of Railroad Standard Time. This artificial creation divided the continental US into four different zones within which time was the same (Daniels, 2000; Stover, 1970). Railroad standard time replaced the over 500 local times that existed in 1883: these exhibited uniqueness across different geographical longitudes because true time was measured using clocks calibrated to the local

position of the sun (Bluedorn, 2002; Daniels, 2002). Eventually, Railroad Standard Time became the *de facto* temporal standard beyond railroads, influencing and organizing private and public life, leading to complaints that people would need to “marry by railroad time”, “die by railroad time”, and that even ministers would have to “preach by railroad time” (Stover, 1970, p. 70). While the system of time zones has become ubiquitous and is rarely remarked upon, it is an early example of a standardization mechanism that enabled coordination.

### *The Design of Work*

The formal study of coordination began shortly after the appearance of railroads, with the emergence of large-scale manufacturing (Scott & Davis, 2007). In the late nineteenth and early twentieth century, manufacturing employment rose dramatically, along with the size of manufacturing establishments (Jacoby, 1984). With increases in scale and speed of production the old “drive system” of management, in which foremen pressured workers to increase output, was no longer an efficient way to organize work. Not only was this system arbitrary and unfair to workers, but foremen were typically bad record keepers and disorganized managers, limiting the potential benefits of such a system (Jacoby, 1984).

The problems inherent in the drive system served as the impetus for the development of schemes to maximize efficiency and better coordinate through the design of work (Chandler, 1962). One of the most famous methods to maximize efficiency was scientific management, which operated by examining the work that was being performed and decomposing it into its most basic elements, thereby allowing for specialization and the reduction or elimination of waste. This approach, perhaps best exemplified by the engineering methods of Frederick Taylor as well as Lillian and Frank Gilbreth, attempted to rationalize work and find the best way of accomplishing it. Their methods focused on the proper design of tools, fixtures, and even the raw materials for manufacturing through standardization and the interchangeability of parts. In addition, scientific management also advocated the precise and rational use of workers within the manufacturing setting (Taylor, 1916).

The implementation of rationalization in manufacturing through specialization, though, also created additional demands for approaches to bring together the inputs from specialized tasks into an integrated output. Scott and Davis (2007) illustrate the consequences that specialization can have when they note that the adoption of specialized (and improved) tools can lead to increased demands for communication. To ensure that the right tools are available in the right place at the right time for the work to be performed, organizations have to improve scheduling and other allocation mechanisms. With the rationalization of manufacturing work through its design, we begin

to see the challenges that specialization imposed by increasing the need for integration activities.

### *The Design of Organizations*

Later scholars took a different perspective on coordination through an emphasis on the design of management systems rather than the design of work. Perhaps the best known management scholar to approach the problem of rationalizing organizations was Henry Fayol, who was inspired by Frederick Taylor (Fayol, 1949). For Fayol, the rationalization of organizations began with proper design. In his view, work could only be accomplished if the right management systems were put in place. Fayol's perspective was guided by principles for hierarchical systems, such as the unity of command, centralization, and the subordination of individual interests. Fayol (1949) argued that, in contrast to Taylor's "bottom-up" approach, his was a "top-down" approach, one that ensured that the central element in administration, unity of command, was respected.

Planning formal elements of organizations also featured prominently in organizational design, where scholars tried to determine how to devise organizations according to the contingencies they faced in their environment. Coordination, particularly across departments and functions, was a key outcome of these models. Several organizational design theories focused on uncertainties rooted in the task or technology such as production technology (Woodward, 1970), task complexity (Perrow, 1967), workflow integration (Hickson, Pugh, & Pheysey, 1969), and task interdependence (Thompson, 1967). These uncertainties influenced the structure of coordination in organizations: with more variable or uncertain technology, organizations relied less on impersonal structures like hierarchy, rules, programs, and more on interpersonal communication or feedback (Van de Ven, Delbecq, & Koenig, 1976; Argote, 1982).

A significant characteristic of the approaches advocated by Taylor, the Gilbreths, Fayol, and organizational design was the deliberate nature of their attempts to plan systems to produce coordinated activity. This deliberate formalization occurred either through the design of work (Taylor, 1916) or the design of relationships between positions in the organization (Fayol, 1949). In essence, coordination resulted from the specification of exchanges between areas of work through roles, rules, and structures (physical and organizational). The central assumption of these approaches was designability: these scholars believed that organizational systems could be articulated with enough specificity and precision to allow for individuals to fully complete the work. Such formalization continues to be seen as an essential feature of organizations, in which the "rules governing behavior are precisely and explicitly formulated", and "roles and role relations are prescribed independently of the personal attributes and relations of individuals" (Scott & Davis, 2007, p. 29).

These theories, heavily reliant on design, were developed in the era in which manufacturing and production work was the dominant form. As a consequence, some of the distinctive characteristics of this context, such as the tangibility of its products, influenced the theorizing and conceptualization of coordination. For example, in most manufacturing operations the product of the work is visible and measurable, so progress towards the task and its completion can be monitored directly. Similarly, the physical nature of a manufactured product means that the process to create it can generally be examined, rationalized, and decomposed into sub-elements which can be integrated to complete the product through the use of assembly lines. The objects of manufacture, themselves, can aid in making the interdependencies between different groups evident. The physical nature of a manufactured product also means that problems can be more easily identified, and thus resolved. Lastly, in early manufacturing settings many of the interdependent parties were co-located, making everyone's work visible, enabling problem resolution across groups.

As the nature of work has changed in organizations due to the shift away from manufacturing, the limitations of these coordination theories have become evident. In many instances, organizations provide services rather than products, the nature of which is less tangible (Davis, 2003). In other instances, such as the creative work involved in product or software design, a single optimal solution may not exist and progress towards completion may be difficult to estimate (Kraut & Streeter, 1995). Moreover, as the boundaries of organizations have become blurred, the difference between what lies at the technical core of the organization and the periphery has become ambiguous (Scott, 2004; Hargadon, 2003). In each of these cases, interdependencies between different pieces of work may be uncertain or challenging to identify, making it difficult to know who should be involved in work and whether there is a correct order in which parties should complete their own specialized work.

Another shortcoming of the earlier coordination literature is the portrayal of processes and structures as formal elements planned by organizations rather than as ongoing work activities that emerge in response to coordination challenges. This limitation is important because formalized, designed, planned, or engineered solutions often fail to account for the many eventualities that organizations face in the performance of their work. In theories of organizational design the informal behavior necessary to deal with such unplanned contingencies was typically collected into a catch-all category labeled "mutual adjustment" (Thompson, 1967), "lateral relations" (Galbraith, 1973), or "ad hoc coordination" (Donaldson, 2001). Because this category was difficult for scholars to measure, these informal and emergent coordination practices in organizations remained largely unexamined. This is unfortunate because unplanned contingencies and the responses to them represent important influences in organizations.

The literature on coordination that has developed in the last two decades has taken great leaps to address these shortcomings in our understanding of coordination. In addition to examining the formal and designed aspects of coordination, contemporary researchers have also devoted considerable attention to what was labeled “informal behavior” in earlier work. By taking into account both the nature of post-industrial work and the emergent nature of coordination, these new conceptions are beginning to remedy the outstanding gaps in our understanding of coordination in organizations.

### **Recent Approaches to Coordination**

Recent research on coordination has extended beyond organizational theory into a variety of disciplines such as computer science, information systems, and sociology of work. Our review of this newer work leads us to agree that “at its core, coordination is about the integration of organizational work under conditions of task interdependence and uncertainty” (Faraj & Xiao, 2006, p. 1156). While new research on coordination shares with earlier research an interest in the tasks carried out by organizations, the approach to analyzing these tasks differs. As we have noted, early work by efficiency experts and organization design theorists rested on the assumption that organizational arrangements can be designed for optimum performance. In contrast, new research on coordination instead explores how the work of such arrangements is executed. A difference in the recent research is that it is less concerned with optimizing structures for a given environment, and instead considers coordination as it happens, assuming that people in organizations must coordinate the work regardless of the organizational design.

The expansion of coordination research into new arenas has engendered a greater range of theoretical and empirical concerns, resulting in new conceptions of and labels for coordination. For example, research in this domain has focused on communicating (Ballard & Seibold, 2003), cooperation (Pinto, Pinto, & Prescott, 1993), knowledge sharing (Bechky, 2003a; Carlile, 2002), and interaction (Heath, Hindmarsh, & Luff, 1999). Moreover, scholars in the recent wave of research have proposed a variety of definitions for the term “coordination” which address their particular theoretical conclusions (Table 10.1). The multiple definitions, while demonstrating some variation due to their range of focal interests, still embody evident commonalities: (1) people work collectively; (2) the work is interdependent; and (3) a goal, task, or piece of work is achieved.

Given the variety of definitions, for the purposes of our review, we will adopt Faraj and Xiao’s (2006, p. 1157) definition of coordination as a “temporally unfolding and contextualized process of input regulation and interaction articulation to realize a collective performance”. This definition best fits the spirit of this recent wave of coordination research, reflecting scholars’ shared interest in the emergent nature of the process of coordination. For instance,

Table 10.1 Selected Definitions of Coordination

| Authors                               | Definition                                                                                                                                                                                                              |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Simon (1945)                          | "The adoption of all members of a group of the same decision" (p. 8)                                                                                                                                                    |
| Van de ven, Delbecq, & Koenig (1976)  | "The integration or linking together of different parts of an organization to accomplish a collective set of tasks" (p. 322)                                                                                            |
| Argote (1982)                         | "Coordination involves fitting together the activities of organization members, and the need for it arises from the interdependent nature of the activities that organization members perform" (p. 423)                 |
| Malone & Crowston (1990)              | "[T]he act of working together harmoniously" (p. 358)                                                                                                                                                                   |
|                                       | "[T]he act of managing interdependencies between activities performed to achieve a goal" (p. 361)                                                                                                                       |
| Bailetti, Callahan, & DiPietro (1994) | "[W]e define coordination structure as the entire set of interrelated interdependencies between all of the individuals and groups in some problem domain" (p. 395)                                                      |
| Kraut & Streeter (1995)               | "Coordination has been defined as the direction of 'individuals' efforts toward achieving common and explicitly recognized goals" (p. 69)                                                                               |
|                                       | "In software development, it means that different people working on a common project agree to a common definition of what they are building, share information, and mesh their activities" (p. 69)                      |
| McGrath, Arrow, & Berdahl (1999)      | "Coordination of interests refers to the functional interconnections between member interests and goals and group interests and goals" (p.1)                                                                            |
|                                       | "Coordination of understandings refers to the development of shared perceptions and meanings among members, including an appreciation of the ways in which members reliably see and interpret events differently" (p.1) |
|                                       | "Coordination of action is a synchronization and sequencing of member actions in time and place" (p.2)                                                                                                                  |
| Heath & Staudenmeyer (2000)           | "Organizing individuals so that their actions are aligned" (p. 154)                                                                                                                                                     |



- Ballard & Seibold (2003)  
“Coordination can be defined as the collective accomplishment of individual goals through a cooperative process” (p. 401)
- Quinn & Dutton (2005)  
“Coordination is the process through which people arrange actions in ways that they believe will enable them to accomplish their goals” (p. 36)
- Faraj & Xiao (2006)  
“At its core, coordination is about the integration of organizational work under conditions of task interdependence and uncertainty” (p. 1156)  
“A temporally unfolding and contextualized process of input regulation and interaction articulation to realize a collective performance” (p. 1157)
- Fussell et al. (1998)  
“The extra work organizations and individuals must complete when individuals are working in concert to accomplish some goal, over and above what they would need to do to accomplish the goal individually” (p. 3)
-

Kellogg, Orlikowski and Yates (2006, p. 28) describe the coordination practices used in an online advertising firm to “facilitate their dynamic, uncertain, and ongoing accommodation to each other and clients”. Similarly, Faraj and Xiao (2006) demonstrate how dialogic coordination, time-sensitive interaction among team members, is used in trauma centers when standard procedures are insufficient to coordinate the work of saving patients. In these cases, coordination is accomplished on the ground as the work progresses. Scholars are also studying cases where organization design is expected to fall short. As Majchrzak, Jarvenpaa, and Hollingshead (2007, p. 147) point out, when disasters strike, formal structures and planned responses are “too slow, disconnected, and inadequate for the task”, and emergent group responses are necessary to coordinate disaster relief. By transcending the assumptions of planned approaches, this work shows that understanding emergent action in context is critical to developing a full picture of coordination.

Our initial goal in our literature review is to bring together the varying perspectives on emergent and contextualized coordination processes. To organize this research, we first focused on the ways in which authors suggested that coordination is enabled or accomplished in the organizations they studied. This, of course, raised some challenges: the development of a contextualized perspective in this research across different disciplines generated not only different definitions, but a messy theoretical picture. We therefore sought to find a way to reconcile the conflicting definitions, bring together the disparate findings, and untangle the theoretical explanations to provide an overarching framework.

Our criticisms of the literature and our ultimate unifying framework will come later. First, however, we will start our assessment by analyzing the ways in which coordination is accomplished and surveying the literature through the lens of the mechanisms that enable coordination: processes such as routines or meetings, as well as artifacts such as schedules and blueprints, and other methods. To do this, we classify the tools, technologies, and interactions that are represented in recent research. We present this simplification as a stepping stone to develop a deeper understanding of how coordination processes operate.

### **Mechanisms for Coordination**

Coordination mechanisms are the organizational arrangements that allow individuals to realize a collective performance. Mintzberg (1989, p. 101) suggests that coordination mechanisms are “the most basic elements of structure” in organizations and include both formal and emergent elements. In the literature, these organizational arrangements often involve tools, technologies, or interactions that bring interdependent elements together. For example, they include arrangements like assembly lines, which bring materials to workers in a defined sequence, and roles, which can define the responsibilities of

interdependent parties. Here, we present five different types of mechanisms that encapsulate how emergent practices assist in coordination: plans and rules, objects and representations, roles, routines, and proximity.

### *Plans and Rules*

Plans and rules, conceptualized as purposive elements of formal organizations, are deemed indispensable to organizing (March & Simon, 1958; Scott & Davis, 2007). In early conceptions of organizations, managers were seen as responsible for the creation of these plans, whether large-scale and strategic or smaller and tactical, after which they were handed down a hierarchy to be implemented by those lower in the organization. Rules are complementary, because they establish relationships between different parts of the organization and serve to guide choices between alternatives when conflicts appear. These perspectives on plans and rules are evident, in particular, in Taylor's (1914) and Fayol's (1949) work. Within the new literature on coordination, plans and rules are presented in a narrower way, with an emphasis on prospective preparation for task completion.

*Defining responsibility for tasks.* Perhaps the most straightforward way in which plans and rules coordinate is by explaining the actions that different parties have to take to complete a task. For example, in trauma centers, coordination takes place through the reliance on protocols and rules (Faraj & Xiao, 2006), which incorporate the knowledge of different experts involved in patient care and establish the sequences of activities for providing it. Pinto et al. (1993), in their examination of project-based teams at hospitals, also noted that some aspects of the work are controlled through plans and rules. These reflect "the degree to which activities or tasks on the project team were mandated or controlled". Plans and rules can be developed at the organizational or team level. Plans and rules act by making it easier for people to relate to each other and to respond to work as it progresses.

*Resource allocation.* Plans and rules are also useful in coordination because they can ameliorate problems that stem from interdependence and be used to match scarce resources to the tasks that require completion (Crownston, 1997). For this purpose, schedules are often used. Schedules place activities on a temporal map, using prearranged points as timing references (Ballard & Seibold, 2003). For example, scheduling in a hospital follows rules to ensure that the same operating room is not scheduled for two simultaneous surgeries (Bardram, 2000), attempting to resolve the very real limitation of using the same space at the same time for two different purposes. By following such rules in the development of schedules, a hospital can ensure both appropriate usage of limited resources and avoid any double-booking of resources that can lead to conflict and disruption.

*Developing agreement.* Plans and rules can also be useful for coordination through the agreement that they engender among parties. Bechky and Okhuy-sen (2009) show that plans, such as schedules and location maps, are created in both a SWAT team and film production crews through an iterative process. The interaction required to create schedules and maps highlights conflicts, discrepancies, and other difficulties in performing the work. The final product embodies a working agreement of how the mission or project will unfold, and serves as an orienting device for the work of the different individuals in these groups. As Pinto et al. (1993) note in project teams, these group-level plans and rules can increase the level of cooperation across individuals in the group.

### *Objects and Representations*

Scholars studying the use of technologies, objects, and other representations show that they are useful for coordinating work activities. Rafaeli and Vilnai-Yafetz (2004) theorize that objects have three functions in organizations—instrumental, symbolic, and aesthetic. Because coordination is primarily an instrumental function in organizations, this property of objects is most often emphasized in the literature. Conceptually, the literature that explains how representations coordinate work draws on the idea of boundary objects from science studies (Star & Griesemer, 1989). Boundary objects “inhabit” several intersecting social worlds and satisfy the information requirement of each of them (Star & Griesemer, 1989, p. 393). Boundary objects can be used across social worlds to convey both technical and social information and mobilize action (Henderson, 1999; Star & Griesemer, 1989) because knowledge and social dynamics inhere in material objects (Winner, 1980; Latour, 1988, 1996). Objects and representations therefore coordinate by providing information. They also offer a common referent around which people interact, align their work, and create shared meaning.

*Direct information sharing.* Representation technologies enable direct information sharing across groups in a variety of settings, from advertising to medicine to engineering and navigation teams. For example, NASA engineers working in a war-room environment use data spreadsheets to share information that is necessary for the work to progress across different teams (Mark, 2002), and teams on naval ships use a variety of technologies, from algorithms to simple logs, to represent navigational problems in ways that make it easier for the sailors to see and understand others’ activities (Hutchins, 1995). Each of these technologies provides a common space for interdependent groups to post information that is vital for the current task. Posting information also makes the activities of the different groups visible, which helps coordinate their future work.

*Scaffolding.* Representations are sometimes created by groups to impose order by providing a scaffolding or structure for activities. People use this

scaffolding as a reminder of which tasks still need to be done, and who needs to do them, in order to complete the work. For example, Kellogg et al. (2006) describe how advertising groups use a “discovery matrix” to represent elements of the task they are working on. This representation provides a common framework that reminds individuals of what they need to do as a part of the larger project. Similarly, Mark (2002) describes how “maps” of interdependencies are used by NASA engineers to anticipate information demands, making it clear who should receive a given piece of information and who should be the one delivering it. In both cases, coordination of the work is enabled through a representation that makes the progress of the task obvious.

*Acknowledging and aligning work.* As both Bechky (2003a, 2003b) and Carlile (2002) demonstrate, boundary objects are useful for solving problems that occur between groups in production organizations, because when people from those groups come together to interact about the work, such objects allow them to acknowledge their progress on the task. For instance, Henderson (1991) shows how drawings and prototypes help design and production engineers advance the introduction of a new turbine engine by mobilizing discussion and action. Moreover, Carlile (2002) demonstrates that engineers use objects to make their interdependencies clearer while designing automobiles. The specialized groups involved in these activities interact around objects, which enable them to align their activities with one another and acknowledge the work that is being done.

*Creating a common perspective.* By facilitating learning across groups, objects and representations can assist in the creation of shared understandings of the work process across organizations. These shared understandings help coordinate the work. For instance, a representation such as a patient protocol helps to make medical practices more concrete, as Faraj and Xiao (2006) point out. By sharing protocols, those involved with treatment develop “a common mental model of patient condition and treatment options” (Faraj & Xiao, 2006, p. 116). Representations can help group members develop a broadly shared view of the work and tasks the entire group will carry out. Bailetti, Callahan and DiPietro (1994) show that in chip design and manufacture, “coordination ensembles”—which are maps of interdependencies between tasks and group members—help create a common view of the project that aids in coordination.

### *Roles*

Scholars have also identified roles as a useful means for coordinating work activity. Roles represent expectations associated with social positions, and therefore can facilitate continuity of behavior over time (Hughes, 1958; Biddle & Thomas, 1966). At the same time, roles can be loosely and dynamically structured, as expectations are negotiated in interaction (Turner, 1986). This

conception of roles as both able to structure expectations and negotiated over time offers a way to understand how roles enable coordination. Understanding the relationship between roles in organizations, or the role structure, helps people acquire a general sense for who does what in the work process (Bechky, 2006). Also, knowing what tasks are associated with particular roles helps coordinate across people, and negotiating behaviors in a role guides people in the organization so they can learn these expectations.

*Monitoring and updating.* Roles represent the relationships between people and structure individuals' interactions with one another. In organizational theory, the use of roles for coordinating behavior through monitoring and updating is one of the traditional functions of a formal hierarchy. When subordinates report their activities to their supervisors, or supervisors check in on their subordinates, they are coordinating the tasks of the group. In ER teams, for instance, the hierarchy between attending, resident and student doctors coordinates the work—the attending physicians monitor how those beneath them progress on the case and interrupt or jump in if the work is not progressing correctly (Klein, Ziegert, Knight, & Xiao, 2006). The overlapping roles of medical professionals in trauma teams also allow for what Faraj and Xiao (2006) call “plug and play teaming”. In these teams professional control, roles, and hierarchy give people the tools to monitor the progress of the tasks and to keep members accountable to one another for their activities, coordinating the work.

*Substitution.* Roles help create shared understandings of the responsibilities for tasks. For example, when individuals understand the tasks linked to each others' roles, they can substitute for one another in task execution. Such substitution can be seen in ship navigation, where teammates such as the plotter and bearing timer-recorder's shared understanding of tasks enables them to fill in for others (Hutchins, 1991). This understanding of others' roles and their tasks are often developed informally. Bechky (2006) highlights that in the film industry individuals early in their careers spend time rotating through different departments. This allows them to develop common understandings of the task requirements of different roles and how they relate to one another. Thus, although film crew members work on many short-term projects with different individuals filling different roles, they have strong shared understandings of the tasks performed by each role that enables immediate coordination from the first day of shooting.

*Creating a common perspective.* Research also shows that particular inter-group roles are important for sharing information about the work process across the organization. For instance, Gittell's (2002) study of a hospital suggests that boundary spanners in groups can provide information about patients across different groups in the organization, which helps to enhance coordination by making clear what tasks remain to be completed. A similar role

is pointed out in Barley's (1996) work on technicians, who often serve as brokers between other occupational groups in organizations, translating information to create a common perspective that helps groups coordinate their tasks.

### *Routines*

As with plans and rules, routines have been a part of conceptions of organizations at least since March and Simon (1958) discussed them. Routines can be defined as "repeated patterns of behaviour that are bound by rules and customs" (Feldman, 2000, p. 611). In early conceptions, routines were sometimes viewed as mechanistic properties of organizations, as stable and difficult to change. Routines have been treated as stores of knowledge (Nelson & Winter, 1982) and as the outcome of habitual behavior (Gersick & Hackman, 1990). More recently, Feldman (2000, 2004) and her colleagues (Feldman & Pentland, 2003; Feldman & Rafaeli, 2002) have conducted research that moves away from simple conceptualizations of routines, focusing instead on the complex ways in which social meaning and social interaction are embedded within them. This interpretive and emergent conception of routines has expanded our understanding of the ways in which they coordinate. Routines coordinate by providing a template for task completion, by bringing people together, and by creating a common perspective across groups.

*Task completion/stability.* Routines help coordinate by making completion of the task visible. In the simplest manner, when they establish a sequence of activities to be performed, routines enact a way for interdependent parties to observe progress on the task. For example, Feldman (2000) notes how a housing office performs its work in cycles lasting a full year. Some of the task activities performed within that office include the selection and hiring of resident advisors and organizing move-in weekends. By having these routines in place, the parties involved in their execution can split the tasks into subtasks and broadly anticipate who will be participating in completing the task at a given time. Importantly, though, the performance of each instance of a routine can be different from the last because these routines are executed once a year, in many cases by different personnel, with changing objectives, and with a constant emphasis on learning and improvement. This means that while it is possible to expect that a given activity will take place or a given objective will be met, it is harder to predict the details of the work performed to complete those activities or achieve the objectives.

*Hand-off work.* Another way in which routines operate as coordinating mechanisms is by establishing how work moves from one group to another to complete an interdependent task. For example, Symons, Long and Ellis (1996) describe how a "procedure trajectory" in a radiology department establishes how to move patients from department to department, and how to track this

movement through reports. Similarly, Gittell (2002) finds that the specification of task sequences through “clinical pathways”, the formal procedures in routine joint replacement surgery, also coordinates action. In an advertising firm, Kellogg et al. (2006) note a similar routine in which workers use a set of procedures to exchange files of their work to revise them for presentations to clients. In each of these cases, a central feature of the mechanism used is the ability to determine when one group has completed its work and another takes over.

*Bringing groups together.* Routines work as coordination mechanisms, in part, by providing moments for interdependent parties to act jointly on a task. For example, Feldman (2000) shows how bringing the athletic department into a university’s move-in routine (by having them provide ice cream to the new students) makes the athletics department more aware of the challenges faced by the housing department. In particular, it makes it evident that if athletic games are scheduled during the move-in weekend, the move-in process for students is made more complicated. Feldman and Rafaeli (2002) argue that as routines create connections among interacting parties in organizations, the parties are guided through their mutual interdependence. The connections that are embodied in the routine, by facilitating interactions among individuals, make progress on the task possible.

*Creating a common perspective.* Routines can also function as coordinating mechanisms as they help establish a common perspective on the work to be done by a group. For example, Edmondson, Bohmer and Pisano (2001) examine how groups build routines for cardiac surgery through training after new technology is introduced into a team. They note that some groups elaborate on the “shared meaning” of the task through reflective practices in the process of training, which lead to better subsequent performance in the operating room. In particular, for these teams understanding the distribution of task knowledge among individuals and how to use the new technologies are critical elements. Similarly, Okhuysen (2005) explains that for police SWAT teams’ group training sessions, in which they rehearse routines to execute work, are a useful mechanism to establish a commonality of perspective towards the task.

### *Proximity*

The final mechanism that supports coordination within organizations is people’s physical proximity to one another. Distance has an influence on the amount of interaction and communication between people in organizations (Allen, 1977). For instance, early studies showed that people’s friendship, marriage choices, and general attraction to others were related to their distance in space (Festinger, Schacter, & Back, 1950; Festinger, 1951; Newcomb, 1956). Within organizations, proximity’s influence on communication and liking is mediated by visibility and familiarity, which impact coordination.



*Visibility.* Proximity, or physical co-location, is the primary means of creating visibility in organizations (although see Wilson, O’Leary, Metiu and Jett (2008) for an interesting analysis of perceived proximity). Through co-presence (Mead, 1934; Goffman, 1963), organization members can see what others working on the task are doing, get immediate evidence of the progress of the work, and adjust their own work accordingly.

*Monitoring.* Visibility is created through formally established monitoring practices as well as informal, lateral noticing of others’ behavior. For instance, visibility is often used to coordinate activities in medicine: trauma room professionals are able to monitor tasks through their physical co-presence (Klein et al., 2006); in research in other hospitals, clinical rounds allow for professionals to update one another about a patient’s condition (Gittell, 2002). Such monitoring is the norm for medical occupations, where senior members of the group oversee the novices. This proximity allows experts to coordinate the work, intervene when necessary, and make sure that treatment progresses appropriately.

In other types of workplaces, physical proximity creates lateral visibility to coordinate the work. Whether performing surgery (Reagans, Argote, & Brooks, 2005), planning NASA missions (Mark, 2002), or building radios (Liang, Moreland, & Argote, 1995), group members’ visibility is enhanced by their co-location, which allows them to easily keep track of what others are doing. Hutchins (1991) also describes the importance of visibility for accountability in navigation teams: members utilize a “horizon of observation”—a clear view of what others are doing—as they jointly pilot the ship. In these cases, workers are not formally supervising one another, but can link their own interdependent tasks by seeing others’ task performances.

*Updating.* Because visibility enables easy updating on task progress, in cases where co-location is difficult, people attempt to make the work visible by communicating the status of the work through other means. For instance, creating visibility is more difficult in large-scale, geographically distributed work such as software development (Metiu, 2006), and thus developers relied on written notes and e-mails to account for the progress of the work. Bechky (2006) points out that in the case of film production, where workers were accustomed to coordinating through physical proximity, when they could not be physically visible they updated one another frequently about their activities via walkie-talkie. Even in cases where surgeons are co-located, when their view of the work process is constrained, they sometimes adopt technological tools that create visibility to account for what happens inside a patient as they work together (Edmondson et al., 2001).

*Familiarity.* Proximity also leads to familiarity (Newcomb, 1956; Allport, 1954), which can enhance coordination. In the broader organizational

literature, familiarity has been defined as the understanding that individuals have of others (Okhuysen, 2001). Familiarity has a cognitive element in the knowledge that individuals have of each other, an emotional element in the form of affect, and a behavioral element which is the action consequence of knowledge and affect (Rockett & Okhuysen, 2002). In the literature on coordination familiarity is used more narrowly, focusing on task-related information that is acquired through direct experience working with others. As a consequence, it tends to be very specific and focus on the knowledge one person has of another particular individual or situation. In general, researchers assume that increased familiarity leads to stronger relationships, which encourages individuals to embrace their interdependence and allow them to more effectively coordinate (Gittell, 2002, p. 1410).

*Anticipating and responding.* Kraut, Fussell, Lerch and Espinosa (2008) found that coordination improves as members work with each other longer. In this case, improvements in coordination included the elimination of the timely transmission of accurate information. Kraut et al. (2008, p. 5) argue that familiarity improves coordination because it allows for “group members to anticipate the actions of others and to make adjustments in their own behavior in anticipation of others’ reactions”. Examining joint replacement surgery teams in a teaching hospital, Reagans et al. (2005) also show that efficiency improves as individuals acquire greater experience working with one another, because as individuals learn about the preferences of colleagues, they can tailor their own actions to their activity partners. Such gains in productivity through familiarity have also been noted in mining crews (Goodman & Leyden, 1991), where tacit coordination improves as individuals gain experience working with each other.

*Store of knowledge.* Familiarity helps with the development of a transactive memory system, where individuals learn what others know and can use the group as a store of information to perform the task (Hollingshead, 1998). Examining groups building radios, Liang et al. (1995) show that when groups train together, the transactive memory system they develop allows them to subsequently produce higher quality radios. Information about what others know can help individuals figure out whom to look for to obtain guidance on the performance of work (Olivera, 2000). Examining an apparel and sporting goods company, Austin (2003) showed how having well-developed transactive memory systems in groups can help them accomplish their goals. The key is that a strong transactive memory system facilitates group members’ identification and use of relevant knowledge, enabling members to generate higher quality solutions but also to operate more smoothly in the completion of the task (Austin, 2003, p. 867).

*Developing trust.* Familiarity can also help coordinate through the impact it has on the development of trust among individuals. The development of trust is important in several ways. In the teams described by Liang et al. (1995) members develop the confidence that others possess the radio-building expertise attributed to them. This trust allows individuals to expect that the target person(s) can deliver a “technically competent role performance” (McEvily, Perrone, & Zaheer, 2003, p. 93). Knowing that others have the required expertise and can perform the tasks they are responsible for ensures that coordinated activity proceeds effectively (McAllister, 1995). Reagans et al. (2005), in their study of joint replacement surgery teams, argue in a complementary manner that trust is important because of the idiosyncratic nature of specific dyadic relationships. Each of these relationships involves “relationship specific logics” that can only emerge as partners share private knowledge, and trust enables and is formed through such sharing (p. 872). When individuals can trust one another to perform their work in the proper manner, coordination is enhanced.

### **Integrating Conditions for Coordination**

As our review of coordination mechanisms shows, we have learned quite a bit in recent years about how coordination is accomplished in organizations. The five mechanisms for coordination—plans and rules, objects and representations, roles, routines, and proximity—encapsulate how emergent action coordinates work in organizations. By using these five mechanisms for coordination as an organizing framework for our review, we begin to provide some coherence to the findings from the current literature.

However, a reliance on mechanisms to describe coordination creates several challenges for developing a theoretical framework for coordination in organizations. One problem is created by the way coordinative action is treated in the literature: the interdisciplinary range of coordination research results in a variety of approaches to the object that is being coordinated. Crowston’s work (Crowston, 1997; Malone & Crowston, 1990, 1994), for instance, approaches coordination as a problem of managing the interdependence between tasks. Thus, he focuses on elaborating the mechanisms that may coordinate tasks in which resources are shared, or where tasks need to be decomposed into sub-tasks. In contrast, Gittell’s work examines relational coordination in groups (2000, 2002, 2004). She is concerned with the quality of the relationships in groups and she therefore measures how communication, shared goals and knowledge, and mutual respect influence task performance. Faraj and Sproull’s (2000) work is representative of the research on coordinating knowledge in organizations. They demonstrate how recognizing the need for and location of expertise, as well as the ability to bring that expertise to bear, improves team performance. Because these scholars and others differ in how they explain coordination in organizations, a comprehensive perspective remains elusive.

Another obstacle to comprehensive understanding of coordination is the embeddedness of the findings in particular contexts. For instance, in Okhuysen's (2005) description of police SWAT teams, the development of routines is integrally tied to the regimented and paramilitary nature of the approach used by officers. While a tight fit between the findings and the context is valuable in establishing the relevance and effectiveness of these mechanisms, findings such as these trade-off generalizability for accuracy (Weick, 1979). As a consequence, it is not immediately obvious how one would draw comparisons or establish more general relationships among mechanisms, actions, and outcomes across studies. Further, the literature exhibits a remarkable array of terms to describe similar mechanisms: procedures in one study resemble plans in another, while in a third, a plan might be labeled a representation. In other instances the difficulty is that a specific tool, such as a schedule, is sometimes treated as a plan and at other times is treated as an object. While all of these labels are appropriate for the contexts in which the mechanisms were studied, this is problematic for an overarching conceptualization of coordination.

A third challenge lies in a lack of explication of the means by which coordination happens: a focus on the "how" behind the mechanisms. In many of the studies we surveyed, the explanation for why mechanisms worked to coordinate action, while present, was not front and center. In some studies, scholars articulated this explanation but did not demonstrate it. For instance, in Goodman and Leyden's (1991) study of coal mining crews, familiarity among crew members is inferred from archival records documenting previous experience working together. Their findings then show that those crews in which individuals are more familiar, as measured through those work records, are also more productive. Critically, in this research it is impossible to determine the actual work practices or interactions among the miners; that is, we do not know the details of why or how familiarity might enhance coordination in these crews.

In other studies, mechanisms are fully described and compared, but the means by which they work to coordinate is not explored. For instance, Crowston's (1997) coordination theory describes alternative mechanisms for managing the interfaces between tasks, actors, and resources. He argues that many of these mechanisms are substitutes for one another; for instance, markets and vertical integration are both ways to manage task assignment. However, the theory does not explain why these might serve as substitutes: what are the characteristics of these mechanisms that enable their successful substitution? While Crowston's (1997) typology is a useful one for understanding what mechanisms work in similar interdependent situations, it falls short of a generalizable theory of coordination. Without making distinctions in how mechanisms work, we cannot theorize about what they truly accomplish in the way of coordinating.

In this final part of our review, we present a framework to explain how coordination mechanisms function. We began thinking about the possibility

for a general framework when we noticed that many of the mechanisms that authors explained seemed to fulfill similar requirements. For example, to successfully accomplish emergent coordination, people needed to be aware of and understand how their work fit with the work of others in the organization. On closer examination, we also observed that these types of requirements appeared frequently across the literature. These requirements, in addition, targeted similar underlying properties. To classify them properly, we engaged in an inductive exercise to uncover which were the most pervasive properties and organize them into a set of general conditions. We began coding the articles, and after an iterative process of comparison and contrast (Glaser & Strauss, 1967), we winnowed our categories to three conditions.

We identified three integrating conditions for coordination: accountability, predictability, and common understanding. These integrating conditions for coordination are the means by which people collectively accomplish their interdependent tasks in the workplace. Each of these conditions addresses some of the demands that the integration of specialized work imposes on the individuals performing the work and resolves some of the uncertainties created by interdependence. Central to our framework is the idea that each condition can be accomplished through a variety of mechanisms. Thus, when individuals respond to the demands of integrating specialized work, they do so by enacting different mechanisms that create the integrating conditions for coordination, drawing from a wide variety of options to achieve them. Below, we describe the three integrating conditions for coordination, and explain how the mechanisms in the coordination literature create them (also see Table 10.2).

### *Accountability*

Accountability addresses the question of who is responsible for specific elements of the task. By creating accountability, organization members make clear where the responsibilities of interdependent parties lie. Traditionally, accountability has been thought of as a way to enact formal authority and organizational standards. For instance, reporting metrics hold people in an organizational hierarchy to standards such as “on time arrivals” (Gittell, 2000) or a given “time to fix the problem” (Orr, 1996). However, the use of accountability as an integrating condition in the coordination literature is broader in both scope and means. By making responsibilities visible, parties become accountable for their own contribution while also making other parties accountable for theirs. Moreover, from this perspective accountability is not only attained through hierarchical authority, as in the traditional coordination literature, but can also be achieved in other ways, such as lateral interactions in meetings or through public status reports. This highlights that accountability can be created through informal and emergent action as well as by formal means. Accountability, because it makes responsibilities clear, contributes to

**Table 10.2** Integrating Conditions for Coordination and their Relationship to Coordination Mechanisms

|                             | Accountability                      |                                          | Predictability                      | Common understanding            |
|-----------------------------|-------------------------------------|------------------------------------------|-------------------------------------|---------------------------------|
| Plans and rules             | Defining responsibilities for tasks | Defining responsibilities for tasks      | Defining responsibilities for tasks | Developing agreement            |
|                             |                                     | Resource allocation                      |                                     |                                 |
| Objects and representations | Scaffolding                         | Scaffolding                              |                                     | Direct information sharing      |
|                             | Acknowledging and aligning work     |                                          |                                     | Creating a common perspective   |
| Roles                       | Monitoring and updating             |                                          |                                     | Substitution                    |
|                             |                                     |                                          |                                     | Creating a common perspective   |
| Routines                    | Hand-off work                       | Task completion/stability                |                                     | Bringing groups together        |
|                             |                                     |                                          |                                     | Developing agreement            |
| Proximity                   | Visibility: monitoring and updating | Familiarity: anticipating and responding |                                     | Familiarity: store of knowledge |
|                             | Familiarity: developing trust       |                                          |                                     |                                 |

“aligning actions” among interdependent parties (Ohrbuch, 1997, p. 463; Mills, 1940; Scott & Lyman, 1968).

While accountability can sometimes be designed and built into the formal structures used to perform work, it is not necessarily accomplished through formal means. Consider two different manufacturing plants: a standard hot-plate assembly line (Hackman, Oldham, Janson, & Purdy, 1975) and a semiconductor equipment new product line (Bechky, 2003a). When building a hot-plate, placing the enclosure requires that the cable and plug already be installed by the assembler up the line. Thus, the assembly line technology provides automatic accountability: the assembler who attaches the cable must install it before the hot-plate reaches a second assembler, who is responsible for installing the enclosure, and who can readily observe if the assembler up the line has completed his or her task. The formal structure of the assembly line makes both assemblers aware of their interdependence with each other and the task responsibility of both parties.

In contrast, semiconductor equipment assemblers work in teams, and in final assembly, individuals build machines from subassemblies to completion (Bechky, 2003a). There is no assembly line that channels accountability, and engineering drawings are often incomplete. Moreover, assemblers are instructed by technicians about the order of assembly, and it is through these discussions around the prototypes that technicians spell out the responsibilities for assemblers. Thus, accountability in this setting is accomplished through emergent dialogue, rather than a formal assembly structure.

In the literature, coordination mechanisms accomplish accountability in different ways. Plans, rules, and objects create links between tasks and those responsible for them, by providing the scaffolding where the identity of those responsible is specified. For example, in film crews (Bechky & Okhuysen, 2009) schedules tie the filming of individual scenes to the different departments that are necessary to complete each scene (e.g., costume, makeup, scenery, electric, and so on). Once the definition of responsibilities takes place through the schedule, departments and department heads can be held responsible for the activities they must complete during the production. This example also highlights that a critical element in the development of accountability as an integrating condition for coordination is a reliance on trust: to the extent that parties can count on a consistent and reliable performance of the work by others (McEvily et al., 2003), establishing the scaffolding is sufficient to enable coordination.

Accountability can also be enacted through roles, routines, and visibility. These mechanisms enable monitoring, updating, and hand-offs between the parties. For example, Symons et al. (1996) give one example of how routines, in the form of “procedure trajectories” for patient care, ensure accountability among parties by defining how and when handoffs between different responsible departments happen. In a different manner, during joint replacement

surgery (Reagans et al., 2005) team members can hold others accountable for their tasks by continually monitoring their work. This monitoring is possible due to the physical proximity in which surgery takes place. When visibility is not an inherent part of the work arrangement, groups sometimes adopt technologies to make the work visible and create accountability. For example, Kellogg et al. (2006, p. 29) describe how differentiated groups in a Web-advertising firm engage in “display practices” in which they post work-in-process online so that everyone can be “kept in the loop” about the progress of the work. Spreadsheets among NASA engineers in the war-room (Mark, 2002) and logs on naval ships (Hutchins, 1995) serve the same purpose. These tools allow groups to post and share information, updating their progress and making their activities visible, creating accountability.

We began this section by defining accountability as the integrating condition that establishes who is responsible for particular aspects of the task. Our review suggests that accountability can be achieved through plans, rules, and objects which act as scaffolding for the responsible parties. In addition, roles, routines, and visibility make it easier to establish hand-offs between parties, clarifying the responsibility that each one has in the overall task. From our perspective, accountability is important because it enables integration by allowing interdependent parties to ascertain the part that everyone plays in the final production of the work.

### *Predictability*

Predictability, the second integrating condition, enables interdependent parties to anticipate subsequent task related activity by knowing what the elements of the task are and when they happen. In other words, predictability involves having a sense for what subtasks make up larger tasks and in what sequence tasks will be performed. Being able to anticipate task related activity allows parties to plan and perform their own work, and is essential for coordinated activity. As Simon (1945, p. 81) notes, planning and organizing are useful not “merely to put each participant in the job he can best fill, but to permit each to form accurate expectations as to what the others are going to do”. Similarly, Rico, Sanchez-Manzanares, Gil and Gibson (2008, p. 164) note that one of the critical elements of implicit coordination is the anticipation of the actions and needs of others. Predictability allows people to fit their own tasks into the whole through anticipation of when others will do their work.

Predictability can be developed and enhanced through both formal and emergent mechanisms. For instance, consider the difficulties that a container manufacturer has in coordinating activities between sales and manufacturing (Lawrence & Lorsch, 1967). When determining production schedules, the parties in this organization rely on the intervention of the formal hierarchy. By escalating the scheduling of manufacturing operations to the highest levels of the organization and awaiting their decision, the parties are assured not



only that a schedule that is organizationally suitable will be created, but also that counterparties will abide by the solution imposed by management's authority, enhancing the predictability of the relationship and the process.

In contrast, consider the situation of police officers in crisis teams (Okhuy-sen, 2005) who must respond collectively and quickly to dangerous suspects and uncertain situations. During rehearsals and briefing meetings, officers place tremendous emphasis on developing common approaches to different types of situations. Thus, they rehearse fast-paced entries into locations until the process is smooth and, before each mission, they discuss the approach until there are no questions about each officer's role. Over time, this familiarity with each others' roles and responsibilities enables officers to confidently anticipate what actions other members will take when faced with specific contingencies during a mission. Like the formal predictability engendered through hierarchy in the container organization, a consequence of building knowledge of one another through emergent interaction is an enhanced level of predictability in the execution of the task (Reagans et al., 2005).

Predictability is built through familiarity because groups can organize their tasks with knowledge of the people participating, using their partners' preferences to guide the performance of the task. The performance of the task can also be guided by plans and objects, mechanisms which help create predictability by defining the tasks that must be accomplished, as well as the timing or order in which they must happen. For example, in hospitals plans are enacted as medical protocols (Faraj & Xiao, 2006) that provide guidance for the care that patients receive and establish the sequences in which patients travel through different departments in the hospital depending on the progress of their treatment. By providing specificity, plans and objects give a more complete picture of task related activity.

By specifying sequences of tasks that must take place, routines can also build predictability as an integrating condition for coordination. For example, the activities in the housing office at a university, such as student move-in and resident advisor selection, are guided by routines that take place every year (Feldman, 2000). Because these tasks are a part of the routines of the housing office, their execution can be expected by other interdependent parties. Even if the details on how they are executed are not evident or clear to other departments, all parties know that these tasks will be completed every year.

As we noted, predictability allows interdependent parties to anticipate subsequent task related activity. Predictability is achieved through plans and objects that define the tasks that must be completed; through familiarity that allows for an understanding of others' preferences around the work on the task; and through routines that establish the tasks that must happen. When high levels of predictability exist, individuals and groups can count on the successful execution of the work of others and perform their own tasks accordingly, enhancing integrating activities.

*Common Understanding*

Common understanding helps coordinate by providing a shared perspective on the whole task and how individuals' work fits within the whole. Common understanding is high when participants in an interdependent activity share knowledge of the work that is to be done, how it is to take place, and the goals and objectives of the work. The literature considers common understanding of three different varieties. One is a common understanding of the task in terms of the specific "actions and strategies necessary to perform a task" (Cannon-Bowers & Salas, 2001). Knowledge of the different parties in an interdependent situation is also often described as having an effect on coordination with one's interaction partners (Reagans et al., 2005). Lastly, knowledge about the broader context in which coordination activities take place, such as knowledge of organizational goals, can keep everyone oriented towards common task outcomes (Pinto et al., 1993).

Like accountability and predictability, common understanding can be created both through formal, planned mechanisms and by emergent interactions in an organization. For instance, common understanding can be embodied in assembly drawings, schedules, or other plans drawn up by the organization. Thus, common understanding can be developed when plans are created by senior managers and handed down a hierarchy to be implemented by those lower in the organization, as we noted earlier, and as Taylor (1914) and Fayol (1949) might recommend. However, more recent work suggests that common understanding can also be developed when plans are the result of a bottom-up approach, in a more emergent fashion. For example, Bechky and Okhuysen (2009) show that schedules and location maps built in both a SWAT team and film production crews by the members of the groups themselves through an iterative process. By interacting to create these plans, common understandings of how the mission or project will unfold emerge, which orient and coordinate the work of the different individuals.

As the previous examples suggest, plans and rules facilitate coordination by creating common understandings of the whole task the organization is pursuing, enabling the parties to orient their interdependent work. A similar orientation to the whole task is provided by familiarity in organizations. When people train together, they become familiar with the ways in which their work fits together. By developing transactive memory systems, they generate a store of knowledge about how different tasks fit into the whole, and who is likely to perform these tasks best (Austin, 2003). Routines act in a similar fashion, by bringing groups together and guiding the understanding of the collective achievement of the whole task. The connections between tasks and people that are embodied in the routine are what make the development of common meaning possible (Feldman & Rafaeli, 2002).

Other mechanisms, such as objects and roles, are more closely linked to creating understandings of particular ways to complete tasks. Because they facilitate learning across groups, objects help groups translate their different understandings tasks to create a common perspective. For instance, Bechky (2003a) shows how engineers and assemblers use prototype machines to create an understanding of the production process. By demonstrating with prototypes, assemblers translate how they would build a part for the engineers, who then have a better understanding of what is necessary when they return to their design activities. Organizational roles such as boundary spanners (Gittell, 2002) and brokers (Barley, 1996) coordinate similarly. By translating the requirements of different groups, some technicians broker information across groups (Barley, 1996), which allows for shared understanding of particular tasks to spread.

While specific roles create common understanding through direct information sharing, roles also facilitate coordination by enabling substitution of people when necessary to complete tasks. For instance, Bechky and Okhuysen (2009) show how SWAT officers and film production crew members often share extensive knowledge of tasks. Thus, when there is a problem, such as an unexpected absence of a specialized camera operator, another camera operator can fill in so the task can be completed.

Common understanding coordinates through interdependent parties developing a shared conception of the activities they are performing. Plans, rules, routines and familiarity create common understanding at the level of the whole interdependent task or work process. In contrast, objects and roles are conducive to the development of fine-grained shared understandings of how the work itself is performed. Common understanding is important as an integrating condition because it enables participants to apply their effort towards a jointly held conception of the work or of the process to complete the work. Providing this common ground allows individuals and groups to integrate their activities with those of other interdependent parties.

#### *Relationships among Integrating Conditions of Coordination*

Our presentation is organized to provide clear definitions and examples of each of the integrating conditions for coordinated activity. However, a strict differentiation of the integrating conditions is somewhat artificial, because as people work interdependently, multiple options for coordinated action emerge. In most situations where coordination is necessary, we see combinations of all three integrating conditions present simultaneously. Importantly, though, work contexts differ with regard to which conditions are in the forefront of coordinating activities. By exploring how the integrating conditions—accountability, predictability, and common understanding—are related to one another, we can tease out the tradeoffs and complementarities among them.

One way in which the integrating conditions interact is by providing support for each other. For example, one of the clearest relationships we noted in the literature is that between predictability and accountability. In general, we found that accountability supports predictability—making relationships, interdependence, and task progress clear means that the parties involved in the work can easily keep track of others' activities, making them predictable, and helping them deliver their own work in concert. The support between accountability and predictability is perhaps most evident if we consider the role of formal hierarchies in coordination, which define the responsibilities and relationships between people and groups and, by creating this accountability, make the work more predictable across groups. Accountability and common understanding can also support one another. Both Edmondson et al. (2001) and Reagans et al. (2005) show that teams that have both high levels of accountability and common understanding perform more effectively. In this case, the two integrating conditions have a seemingly additive effect.

When we consider the relationship between predictability, the ease of anticipating subsequent task activities, and common understanding, the shared conception of those activities, we can see that integrating conditions sometimes operate as substitutes for one another. One example of how this works is provided by the use of shipping containers to transport goods as one means of creating predictability. Through the use of containers port operators reduce the ambiguity related to the types of equipment required and the process of loading a ship, because all containers are of standard sizes. Such standardization reduces the need to develop common understandings between the ship captain, port operator, truck drivers, and customers—and many elements of the shipping process can be taken for granted. When a customer needs to transport a non-standard shipment, however, predictability becomes problematic because common understanding (otherwise reflected in standard size containers) is missing. For example, an auction house recently shipped a large Koons sculpture from Dallas to London and faced coordination challenges due to the inability to use a standard shipping process ("Splendor in the Grass", 2008). Instead, due to the difficulties of transporting the sculpture (including its size and weight and the desired arrival date), a custom container had to be built, a specialized airplane that opened in the nose was chartered, and the shipment was accompanied across Europe by a police escort. Obviously, the coordination of this shipment could not rely on the use of standards as a mechanism to create predictability. The parties involved instead had to actively develop common understandings about the process of shipping such a unique item. This example shows that in some situations, the relationship between these integrating conditions is one of substitution—common understandings can replace predictability and *vice versa*. In general, each of these examples shows how the integrating conditions that we propose act in concert to enable coordination.

## Discussion

The integrative conditions for coordination we propose are an alternative way to understand the effects of the five coordination mechanisms. By describing how the coordination mechanisms affect interdependent parties through the creation of accountability, predictability, and common understanding, we advance a conceptual perspective that captures both emergent and formal processes of integration. In this section we summarize our integrative perspective before developing the implications of adopting this perspective. In particular, we discuss how an integrative perspective puts a focus on the work being performed. We also explain the need to treat coordination as an ongoing accomplishment by examining the relationship between coordination and other common organizational dynamics. Finally, we speculate on the challenges that new forms of organizing bring to processes of integration.

### *An Integrative Perspective on Coordination*

Our review of the current literature allows us to see the importance that processes of integration play in coordination activities. Using that as a starting point, our perspective characterizing coordination in organizations goes beyond the mechanisms that produce coordination to one that puts special emphasis on the elements that enable integration. Coordination is enabled when the interdependence among parties, their responsibilities, and the progress on the task are all made visible through accountability. Additionally, coordination relies on the ability of interdependent parties to anticipate subsequent task related activity, that is, predictability. Finally, a shared conception of activities and how they are performed, or common understanding, also enables coordination. Through coordination mechanisms that produce these conditions, people succeed in integrating interdependent tasks in organizations.

Critically, earlier work on coordination can also be viewed through this lens, suggesting the broad applicability of our perspective. For example, Fayol's (1949) work on the design of managerial hierarchies within organizations can be seen as an effort to maximize accountability. By establishing authority, reporting lines, and unity of command, the design of hierarchies ensures that interdependencies and conflicts between different areas of the organization can be managed and given resolution, exemplifying the power of accountability for coordination. Similarly, Taylor's (1914) scientific management, with its emphasis on standardizing work, tools, and parts is an extreme instance of establishing predictability through material means. In this case coordination is enabled by reducing ambiguity and specifying, in extensive detail, how work is performed. Lastly, the establishment of Railroad Standard Time in 1883 is a radical example of the development of common understanding. Originally conceived as a timekeeping system for the railroad system to

coordinate its activities, this innovation eventually permeated every aspect of society, establishing one of the broadest forms of common understanding.

Our perspective on integration is also useful for framing the recent literature on coordination in organizations. As we have described previously, recent studies have focused extensively on emergent action. Such action is more readily understood by adopting an integrative perspective that can help us understand interaction and interdependence within the context of the task. Of course, the emphasis on emergent action is not completely new. In fact, every study in the organizational design tradition contained an “informal interaction” category, so we know that these scholars were aware of the emergent characteristics of coordination. However, by getting closer to the work and tasks in the organization, more recent work transforms emergent coordination from an informal and marginal activity to one that is central to explanations of coordination, providing a more complete and accurate picture of the process. As a consequence, we believe that the framework we are proposing applies equally well to the settings studied in the past as it does to the studies reviewed here.

Our work highlights the three integrating conditions that, working together and separately, assist in the enactment of coordinated activity. The adoption of the three integrating conditions as intermediate constructs between coordination mechanisms and coordinated activity is helpful because it provides us with an overarching framework that can explain many of the findings of extant work, providing a richer and yet more parsimonious explanation. The use of the three integrating conditions provides a way to embrace the wide variety of coordination mechanisms that have been identified in the literature by explicitly focusing on what those mechanisms accomplish. In other words, rather than relying on the presence or absence of a particular mechanism to explain how coordination occurs, our framework forces us and others to explicitly consider how people use partial mechanisms to perform the work, enriching our explanations of coordination. Additionally, by focusing on the integrating conditions we grant an important theoretical position to informal and emergent activities, which previously have been acknowledged only as the messy remainder of planned coordination. By explaining how the integrating conditions can sometimes support one another and sometimes substitute for one another, our work also emphasizes the role of the work context in determining how coordinated activity unfolds. The emphasis on context also provides an explanation for the inherent limitations of the grand plans of organizational design perspectives which are unable to articulate fully the requirements of the work.

Using an integrative framework to explore coordination forces attention to the work as it is performed collectively. Moreover, a focus on integration is also consistent with research exploring work in organizations which considers how people interact with one another as well as how their work is situated in

and is shaped by organizational structures (Suchman, 1987). For instance, Friedson (1976) argues that the division of labor is best characterized as a process of social interaction, in which individuals and groups “in the course of coping with the varying circumstances and situations of their work, in some sense [shape] the terms, conditions, and content of their work no matter what the formal mode of organization” (p. 310). Sociologists doing workplace studies in a variety of settings from air traffic control to teleservice work to the London Underground have demonstrated the importance of emergent, situated work activities in how the workplace is ordered (Pentland, 1992; Vaughan, 2008; Heath et al. (1999). The strength of this approach is that it is inclusive in its treatment of coordination activities, and the explanations that result from this approach are strengthened by the incorporation of planned and unplanned activities and their interactions.

### *Coordination as an Ongoing Accomplishment*

By separating the integrating conditions from the coordination mechanisms and the realization of coordination, we also highlight the ways in which coordination is an ongoing accomplishment in organizations. The integrating conditions of coordination are necessary, but are not always sufficient for coordination to occur. Normal, everyday dynamics in organizations intrude on coordination by eroding the integrating conditions. In these cases, the integrating conditions need to be reestablished for successful coordination to occur.

One example of a regular process in organizational life that impacts coordination is turnover (Reagans et al., 2005). As individuals come and go in organizations, coordination can suffer because both new and old members are less knowledgeable about each other, directly affecting the familiarity that enables predictability. In addition, new members also may not be as knowledgeable about the organization’s task—they may be specialists, rather than generalists, for instance (Feldman, 2000)—which reduces common understanding. When turnover occurs, organization members have to find ways to bring individuals “up to speed” on how to work together, for example through training or socialization, so that the task can continue to advance. Alternatively, when turnover is an ordinary and expected occurrence, as in trauma teams (Klein et al., 2006), arrangements that compensate for the expected turnover, such as those that rely more strongly on formalized roles, can be effective at overcoming this challenge.

Another fairly common occurrence in organizations that causes coordination to break down is a mistake in execution. When work is interdependent, errors or mistakes create obstacles for people to accomplish their task. For example, Bechky (2003b) shows how assemblers and technicians cannot rely on the work progressing when mistakes occur in the assembly of semiconductor manufacturing equipment. If particular groups are mistake prone and they are judged unreliable, then there is an erosion of trust and the predictability of



their work is in question (Bechky, 2003b). These mistakes cause a disruption in the production processes that also affects accountability. When accountability is reduced, interdependent parties cannot easily perform their own work, as they lose some of their ability to gauge the progress of the task. Reestablishing that accountability through the active management of problem resolution then allows coordinated activity to continue.

Another ordinary organizational dynamic that erodes the integrating conditions and cause coordination breakdowns are status differences across groups, because high-status groups may feel no need to recognize the task contributions of members of low-status groups. For instance, Metiu (2006) shows how, in software production, high-status programmers refuse to read the notes and comments that low-status programmers provide to document the progress of the work. Because accountability requires an acknowledgement of mutual responsibilities, status differences that prevent such acknowledgement limit the development of accountability. Additionally, in situations where large status differences cannot be overcome, members of groups may not be able to trust one another. When working interdependently, low status individuals in these situations will ask fewer questions and give less feedback, for fear of offending others and the possible repercussions (Edmondson et al., 2001; Lee, 1997, 2002). Such a situation leads to less sharing of knowledge, limiting common understanding in the group.

As these examples suggest, coordination is under persistent attack by the regular dynamics of organizations. What these examples show is that individuals and groups in organizations must constantly recreate the integrating conditions of coordination in order to jointly execute their work. This ongoing accomplishment of coordination relies on emergent action creating the integrating conditions to bring collective and interdependent work together. Additional research is needed to flesh out our understanding of how the integrating conditions and additional dynamic elements of organizations interact.

#### *Directions for Future Research*

Our examination of coordination has a very explicit emphasis on processes of integration, derived from the extant literature. Although coordination also involves processes such as specialization, recent work on coordination has had a dominant focus on the challenges that integration presents. Such an emphasis on integration is not new. It is also evident, for example, in earlier work by organizational design scholars (e.g., Lawrence & Lorsch, 1967). And while this emphasis was lost in the years of empirical testing of organizational design theory, which focused primarily on relationships between environments and organizations (Padgett, 1992), the current wave of research on coordination has returned to a concern with how integration takes place.

However, while much of the current work describes processes of integration in coordination, this emphasis is rarely made explicit. A focus on the



mechanisms of coordination has meant that the development of theory about integration and integration processes is typically embedded in discussions of how the mechanisms operate, rather than receiving attention on its own. Our review, which highlights the conditions that are necessary for integration activity to take place, forces us to separate the consideration of mechanisms from the coordination they help achieve. This theoretical separation moves us towards explaining the effect of these mechanisms on how work is performed, and closer to the development of a theory of integration.

A more detailed and theoretically driven investigation of integration raises important questions, particularly given the changing nature of organizations. For example, new ways of organizing have changed both our understanding of the impact of organizational boundaries as well as our conceptions of work arrangements. “Boundaryless” organizations (Arthur, 1994; Arthur & Rousseau, 1996) have shifted the nature and location of the task boundary. In these organizations, some tasks are outsourced either to individual contractors or to firms. But, in the absence of centralized planning or hierarchical structures, how do interdependent parties create accountability? The research reviewed here suggests some possibilities of emergent forms of accountability that could transcend organizational boundaries and hierarchies. For instance, work that is outsourced to different organizations could be made accountable through display practices such as the ones seen by Kellogg et al. (2006). Additionally, because accountability cannot be created through hierarchy, managers facing these integration challenges might consider emergent forms of influence as an alternative (Kanter, 1989). Another approach could emphasize common understanding instead of accountability by transforming outsiders into insiders by sharing information about operational aspects or making the project a joint endeavor (Hudson & Okhuysen, 2009).

Another boundary that organizations are blurring with increasing frequency is that between the provider and the customer. Here, the challenge is that organizations would like their goods and services to be co-produced by workers and customers to achieve higher levels of satisfaction and efficiency, presumably increasing loyalty. Starbucks’ menu, with its “tall, grande, venti” labels, McDonald’s value meals, and airlines’ self check-in kiosks all require integration of the customer with the organization, such that customers become co-producers. When the customer does not know or follow the script set out by these organizations to create the service, though, they can disrupt the work and impact the quality of service not just for themselves, but for others also waiting in line. These organizations therefore face the need to create common understanding with their customers, who would traditionally be considered outsiders to the organization, in order to integrate the different contributions into the task.

New organizational forms which operate completely outside of traditional boundaries also create challenges for integration. For instance, the work of

open-source software production is carried out by communities of programmers who may be unaffiliated with any organization. This type of work presents a broad predictability problem. While communities have little trouble coordinating the actual coding of software without organizational structures, they face considerable difficulty collaborating with any other entity (O'Mahony & Bechky, 2008). Without a means of representation and governance, potential partner organizations are unable to establish communication, sign a contract, or figure out when the product will be released by the community. In short, they are unable to create predictability. These examples suggest that post-industrial work requires assembling specialized knowledge in ways that we have not done before while facing new task environments. We have speculated here about how coordination might work in these environments, but clearly this is an area that is ripe for future research.

We began this review by noting the increased research activity and interest in coordination over the past two decades. The resulting body of research has expanded our understanding in useful ways. In particular, by approaching coordination through a focus on work this new research explains how formal and informal elements contribute to coordination. This perspective legitimizes the informal elements of coordination that earlier research, such as that focused on organizational design, set aside. The research on coordination in organizations also yields, through our examination, an integrative perspective on coordination. This perspective is useful in deepening our understanding of what happens when work is coordinated, and what it takes for the successful execution of coordinated work. As new forms of work continue to emerge, focusing on the integrating conditions of coordination will provide a robust conceptualization to explain the activities in organizations pursuing these tasks.

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